

Quiz 7 Reflection

SCIE 001 Physics

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1. Neutron Beam Double-Slit

1.1. Specify Question

Question 8: Neutron Beam Double-Slit

You are reading a paper describing a neutron beam double-slit experiment. The experimental setup is as shown below in Figure 1, and the results are in Figure 2.

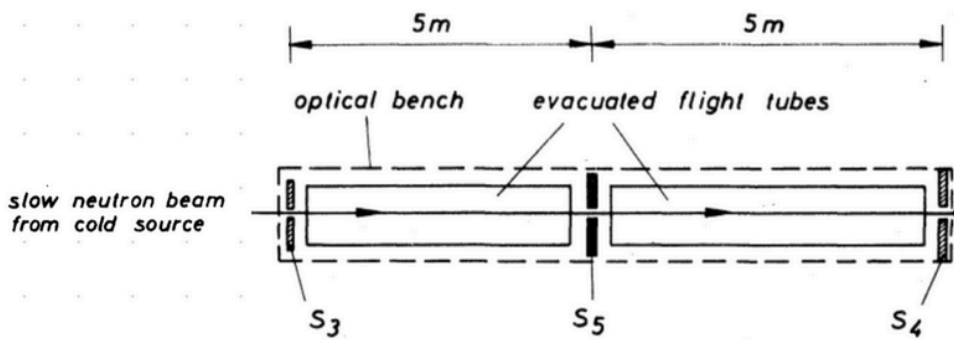


Figure 1: A collimated "slow" neutron beam enters the setup at **S3** and hits the double-slits at **S5**. The double-slits are $112\ \mu\text{m}$ apart. The interference pattern is recorded at **S4**.

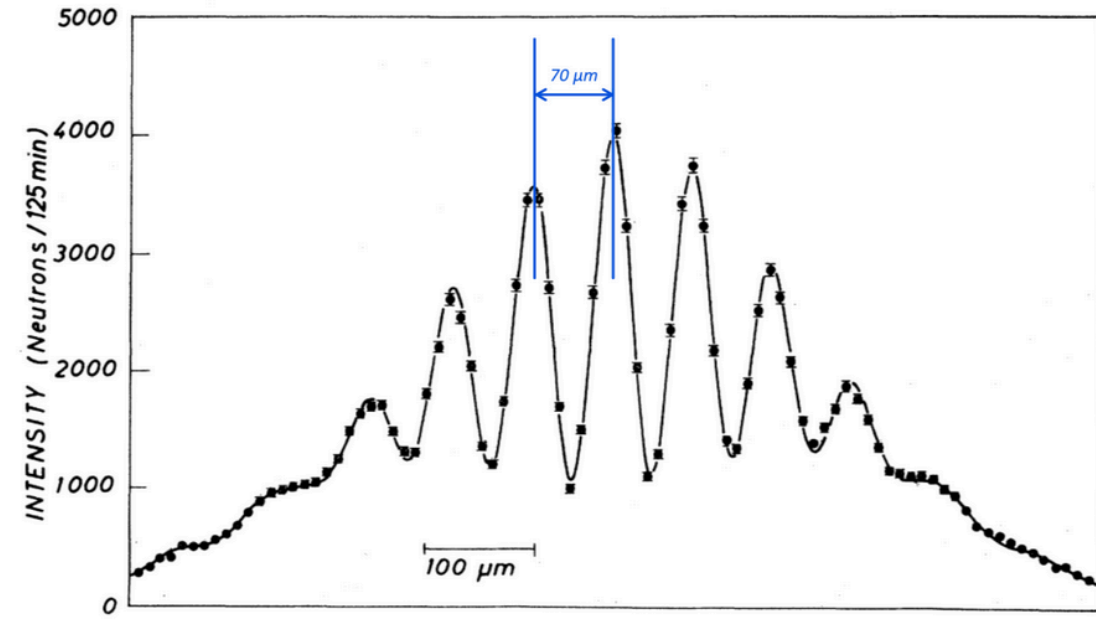


Figure 2: Interference pattern recorded at **S4**.

What was the approximate kinetic energy of these "slow" neutrons? Give your answer in μeV to the nearest integer.

▼ Useful constants

It's convenient to work in eV for this question. Recall that $hc = 1240\ \text{eV} \cdot \text{nm}$ and the mass of a neutron can be expressed as $939.6\ \text{MeV}/c^2$. Alternatively, you can take the mass of the neutron as $1.675 \times 10^{-27}\ \text{kg}$ and work in Joules; in which case, recall that $1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$ and that $h = 6.626 \times 10^{-34}\ \text{J} \cdot \text{s}$

$E_k =$

40

μeV

?

✖ 0%

Correct answer

$E_k = 333\ \mu\text{eV}$

1.2. Diagnosis Phase

During the test I forgot the conversion factor from μm to m and I didn't have enough time to find it in the textbook.

This caused me to panic, as, with both ways of analyzing the question, you had to convert to, or from, micrometers.

Initially, I guessed 340 eV, as was the result of the calculation of the wavelength of the particle if you replaced micrometers with electron volts. However, I changed my answer to 40 eV, as 340 eV looked like too much energy.

Words: 80

1.3. Correction Phase

Using our small-angle approximation, we can reach an expression for fringe spacing in the double slit experiment (given to us on the slides):

$$\Delta y = y_{n+1} - y_n = \frac{\lambda L}{d}$$

Where d is the distance between the slits, L is the length from the slits to the screen, and Δy is the distance between the fringes.

Solving for λ :

$$\frac{d\Delta y}{L} = \lambda$$

We can combine this with our equation for momentum:

$$p = \frac{h}{\lambda}$$

$$p = \frac{h}{\frac{d\Delta y}{L}}$$
$$p = \frac{hL}{d\Delta y}$$

Using momentum-energy equivalence we get:

$$E = \frac{p^2}{2m}$$

Where E is kinetic energy, p is momentum, and m is mass.

Combining:

$$E = \frac{\left(\frac{hL}{d\Delta y}\right)^2}{2m}$$
$$E = \frac{h^2 L^2}{2m d^2 \Delta y^2}$$

To make the math easier, we can multiply both sides of the fraction by c^2 .

$$E = \frac{h^2 c^2}{2m d^2 \Delta y^2 c^2}$$
$$E = \frac{(hc)^2 L^2}{2m (c^2) d^2 \Delta y^2}$$

Now we can substitute, being careful about our units:

$$hc = 1240\ \text{eV} \cdot \text{nm}$$
$$1\ \text{MeV} = 1 \times 10^6\ \text{eV}$$
$$m = 939.6 \frac{\text{MeV}}{c^2} = 939.6 \times 10^6 \frac{\text{eV}}{c^2}$$
$$L = 5\ \text{m}$$
$$\Delta y = 70\ \mu\text{m}$$
$$d = 112\ \mu\text{m}$$

Therefore:

$$E = \frac{(1240\ \text{eV} \cdot \text{nm})^2 (5\ \text{m})^2}{2 \left(\frac{939.6 \times 10^6\ \text{eV}}{c^2}\right) c^2 (112\ \mu\text{m})^2 (70\ \mu\text{m})^2}$$
$$E = \frac{(1,537,600\ \text{eV}^2 \cdot \text{nm}^2) (25\ \text{m}^2)}{2 (939.6 \times 10^6\ \text{eV}) (61,465,600\ \mu\text{m}^4)}$$

With conversions:

$$1\ \text{nm} = 1 \times 10^{-9}\ \text{m}$$
$$1\text{nm}^2 = 1 \times 10^{-18}\ \text{m}^2$$
$$1\ \mu\text{m} = 1 \times 10^{-6}\ \text{m}$$
$$1\ \mu\text{m}^4 = 1 \times 10^{-24}\ \text{m}^4$$
$$E = \frac{(1,537,600\ \text{eV}^2) (1 \times 10^{-18}\text{m}^2) (25\ \text{m}^2)}{2 (939.6 \times 10^6\ \text{eV}) (61,465,600 \times 10^{-24}\ \text{m}^4)}$$
$$E = 0.0003327961166\ \text{eV}$$
$$1\ \text{eV} = 1 \times 10^6\ \mu\text{eV}$$
$$E = 332.7961166\ \mu\text{eV}$$
$$\therefore E = 333\ \mu\text{eV}$$

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